SARATHKUMAR DUNNAPAPAGARI

Aspiring Cloud & ML Engineer · M.Tech CSE Student · B.Tech IT Graduate

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PROFILE SUMMARY

M.Tech Computer Science student (CGPA 8.3) and B.Tech IT graduate with four internships in AWS cloud, Python and data science / ML. Built real-time computer vision, machine learning and full-stack projects with Python, PyTorch, YOLOv8, FastAPI and React. Eager to start my career in a full-time role and build reliable, scalable solutions.

TECHNICAL SKILLS

Languages	Python · Java · C
Cloud	AWS EC2 · S3 · IAM · Google Cloud (GCP)
ML & Data	PyTorch · scikit-learn · YOLOv8 · OpenCV · Pandas · DuckDB
Web & APIs	FastAPI · React · HTML · CSS · JavaScript
Tools	Docker · Git & GitHub · Prometheus · Grafana · Linux · MS Office · AutoCAD

PROJECTS

Construction Site Safety Monitor GitHub · Project page — **Computer Vision · Cloud**

- Built a real-time YOLOv8 pipeline for webcam, CCTV or video files that streams live detections to a React dashboard.
- Designed PPE-ready alerts: helmet / vest violations with S3 snapshots switch on once a custom PPE model is configured.
- Containerized with Docker Compose and NGINX, with an EC2 setup script and Prometheus + Grafana monitoring.

Stack: Python · YOLOv8 · OpenCV · FastAPI · React + TypeScript · Docker · AWS S3 · Prometheus · Grafana

Data Analyst + ML Agent GitHub — **Full-stack · ML**

- Built an AI-assisted analysis platform: charts, plain-English or SQL queries, and 11 ML algorithms (5 classification, 6 regression).
- Added explainability, PDF reports, optional LLM explanations and 14 automated backend tests; works fully without an API key.

Stack: FastAPI · Pandas · DuckDB · scikit-learn · React · TypeScript · Tailwind CSS

EmoMusic GitHub — **Deep Learning**

- Created a music recommender driven by real-time facial emotion recognition that eases the listener toward a calmer mood.
- Fine-tuned EfficientNet-B0 on FER-2013 for 7 emotions, benchmarked against a baseline CNN, with smoothed predictions.

Stack: PyTorch · timm · OpenCV

EDUCATION

M.Tech, Computer Science and Engineering GMR Institute of Technology, Rajam **CGPA 8.3 (till Sem 2)** — **2025 – 2027**

B.Tech, Information Technology Sir C.R. Reddy College of Engineering, Eluru **CGPA 7.29** — **2021 – 2024**

Diploma, Civil Engineering M.R.A.G.R Government Polytechnic, Vizianagaram **67.69%** — **2017 – 2020**

SSC (Class X) Z.P. High School, Amalapadu **CGPA 7.5** — **2017**

INTERNSHIPS

Data Science, ML & AI Intern Datavalley India, with APSCHE *B.Tech long-term internship* — **Nov 2023 – Apr 2024**

- Worked on predictive modeling, applying ML algorithms and improving accuracy through data preprocessing.

AWS Cloud Intern BrainOVision Solutions (AICTE) — **May 2023 – Jul 2023**

- Deployed apps on AWS EC2, managed S3 storage, explored IAM roles, and set up and monitored cloud infrastructure.

Python Developer Intern BrainOVision Solutions — **Jun 2022 – Aug 2022**

- Developed Python automation scripts and small applications, improving efficiency with optimized code.

Industrial Training APSPHC — **Nov 2019 – Apr 2020**

- Gained exposure to infrastructure and technical workflows in public sector projects.

CERTIFICATIONS

- ✓ **NPTEL Cloud Computing** IIT Kharagpur · Elite, 63%
- ✓ **Hands-on Approach to AI for Real-World Applications** IIT KGP
- ✓ **Getting Started with AWS** SkillUp
- ✓ **NPTEL Google Cloud Computing Foundations** 62%
- ✓ **Deep Learning for Developers** Infosys Springboard
- ✓ **FDP on Cloud Infrastructure (AWS)** BrainOVision & AICTE